When Competitors Become Guides: Multimarket Contact and Export Market Entry

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Abstract

This paper studies how familiarity among exporters formed through shared activity in foreign markets affects export market entry. I construct a firm–country–year measure of multimarket contact (MMC) capturing overlap between prospective and incumbent exporters across other destinations. Using transaction-level U.S. Census data on rechargeable battery exports from 2016–2019, I embed MMC in a reduced-form gravity framework to study export entry and subsequent export values. Higher MMC significantly increases both the probability of entry and export intensity, with magnitudes comparable to standard gravity variables.

Keyword: Export entry; Multimarket contact; Gravity

JEL Classification: F12, F14, L13

* Email: boohaker07@gmail.com. Any opinions and conclusions expressed herein are those of the author and do not represent the views of the US Census Bureau or the U.S. Government at large. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7532452: CBDRB-FY25-CED006-014). I am an independent researcher with interests in international trade and applied game theory. I currently work as an economist for a U.S. federal agency. I received my PhD in economics from American University with a focus in international trade and industrial organization. I wrote this paper in my personal capacity. All thoughts expressed in this paper are my own and do not reflect those of the U.S. government. I was a part-time Census Bureau employee and a graduate student at American University when I started this research.

1 Introduction

A large literature documents that firms learn to export from nearby or industry-related peers at home.[1] Much less attention has been paid to learning and interaction that could occur *abroad*, when firms repeatedly encounter one another while serving foreign markets. This paper studies whether such familiarity, formed through multimarket contact (MMC) in foreign markets, affects export market entry and expansion. Given high search costs in differentiated product markets, observing competitors' exports can signal that entering a new foreign market is likely to be profitable (Rauch, 1999; Fernandes and Tang, 2014; Chaney, 2014).

I measure MMC as the extent to which a prospective exporter overlaps with incumbent exporters in other foreign destinations. Using U.S. Census transaction-level data on rechargeable battery exports, I show that firms are significantly more likely to enter a market, and to export more once there, when they have greater multimarket contact with incumbents. Omitting MMC may also bias standard gravity estimates: the coefficients on distance and trade volume, common extensive-margin variables of interest (e.g., Chaney (2014)), shift considerably once MMC is included. Boohaker (2025), motivated by litigation finding evidence of collusion in foreign markets, recently found that higher levels of MMC significantly associated with higher import prices for U.S. rechargeable battery exporters. A vast industrial organization literature has found that MMC significantly corresponds with higher prices and more collusive market outcomes generally.[2] While multimarket contact is often associated with entry deterrence through mutual forbearance (Bernheim and Whinston, 1990; Feinberg, 2013), recent work emphasizes learning, coordination, and information spillovers among competitors.

The next section introduces the mechanism on how MMC affects extensive market decisions. Section 3 discuess the underlying data. Section 4 describes the empirical strategy to show the relationship between MMC and the extensive and intensive margins of trade. Section 5 discusses the results before concluding in Section 6.

2 Theoretical Intuition

The extent of MMC can either promote or discourage entry. Greater overlap in markets can lower fixed costs through learning and network spillovers while also creating opportunities to maintain collusive outcomes across markets. On the other hand, prospective exporters may fear that entry can trigger retaliation in other markets that could create an additional barrier discouraging entry. If the gains from those spillovers and higher markups arising from more market overlap dominate the potential costs of retaliation, then higher levels of MMC with incumbents will increase likelihood of future export entry.

[1] Empirical studies on geographic proximity include Koenig (2009); Koenig et al. (2010); Poncet and Mayneris (2013); Cassey and Schmeiser (2012); Kamal and Sundaram (2016); Fernandes and Tang (2014) and ones on shared industry affiliation include Hong and Wu (2018); Arguello et al. (2020); Alvarez and López (2008); Görg and Greenaway (2004); Zhang and Malikov (2023).

[2] See Yu and Cannella (2013) for a survey. Notable studies include Evans and Kessides (1994), Ciliberto and Williams (2014), Byford and Gans (2014), Jans and Rosenbaum (1997), Parker and Röller (1997) and Feinberg (1985) among many others.

Exporters likely interact with one another and with intermediaries in foreign markets through informal learning, network referrals, and formal partnerships such as joint ventures with local distributors, which reduce uncertainty and help share market knowledge. Trade missions, industry associations, and export promotion events let firms exchange information on demand conditions, quality standards, and distribution networks (Brache and Felzensztein, 2019; tra, 2025b; int, 2025a). Firms producing similar goods may even engage in coopetition, for example by jointly approaching foreign distributors (Liu et al., 2021), and relationships between buyers and sellers can determine a significant portion of trade (Monarch and Schmidt-Eisenlohr, 2023). The industrial organization literature on multimarket contact (MMC), the extent to which competing firms interact across multiple markets, offers a framework for capturing this familiarity. It uses firm–market measures of MMC to test whether contact across markets leads to more collusive outcomes, such as higher prices, greater concentration, or reduced entry.[3] Applying this approach to firm–country trade data lets us examine how familiarity from operating in the same foreign markets affects export decisions, shedding light on firm-level gravity determinants of trade (Chaney, 2014, 2018) and offering insights for policymakers and export-seeking firms.

3 Data

I use the Longitudinal Foreign Trade Transactions Database (LFTTD) to study how MMC between firms effects export entry decisions. The database is compiled and maintained by the U.S. Census Bureau.[4] It records the universe of U.S. international trade transactions. Export data are derived from Export Declaration Form 7525-V filed by exporting establishments and are observed at the transaction level. Each record includes detailed information on the shipped product (classified at up to the 10-digit Schedule B level), shipment value and quantity, whether the transaction occurs between related parties (e.g., within multinational firms), and the destination of the shipment at the port level.[5] I subset this database to identify the set of firms with export transactions of with a Schedule B code starting with 8507 (rechargeable batteries). I then create a dataset of all possible firm-country pairs in each year to study the extensive margin. For the intensive margin, I only keep firm-country pairs if the firm exports to that country in the given year. I study these firms from 2016 - 2018, a relatively recent time period and before the COVID shock, which could introduce unnecessary noise in identifying the effect of MMC in the econometric approach.

I combine this data with two publicly available datasets from the USITC to capture bilateral import flows and distance between any pairs of countries. The first dataset is the, the USITC International Trade and Production Database for Estimation (ITPD-E) which consists of a harmonized panel of bilateral international and domestic trade flows by country, industry, and year, designed for structural trade estimation. It contains variables identifying

[3]Well-known studies include Evans and Kessides (1994), Ciliberto and Williams (2014), and Parker and Röller (1997); see Yu and Cannella (2013) for a synthesis. These studies are motivated by Bernheim and Whinston (1990), who show that multimarket contact can sustain collusion by facilitating punishment and cooperation across markets, even when firms compete non-cooperatively within each market.

[4]See Kamal and Ouyang (2020) for a detailed description of the database construction.

[5]Schedule B codes are product classifications used by U.S. exporters to distinguish among different products shipped to foreign destinations.

exporters and importers, industry classifications, years, and trade values measured in current U.S. dollars, with domestic trade captured when exporter and importer coincide. The database includes flags indicating mirror-reported flows and zero or imputed trade values, facilitating data quality assessment.[6] The second dataset is the USITC gravity database which contains several measurements of trade frictions between any two pair of countries, including the geographic distance between countries.[7] Combining this data with the LFTTD allows one to account for whether the export destinations for a firm factor into their probability of entering a new market in the next period.[8].

4 Empirical Framework

To study export entry, I estimate the following reduced-form model:

$$\begin{aligned}\Pr(E_{ic,t+1}=1) = \Lambda\Big(&\alpha_t + \beta_1 \log MMC_{ict} + \beta_2|C_{it}| + \beta_3 N_{ct} \\ &+ \beta_4 d_{ict} + \beta_5 b_{ict} + \beta_6 d_{US,c} + \beta_7 b_{US,c} \\ &+ \beta_8 \bar{d}_c + \beta_9 \bar{b}_{ct} + \beta_{10} X_{it}\Big), \end{aligned} \tag{1}$$

where $E_{ic,t+1}$ indicates whether firm i exports to country c in period $t+1$. The main variable of interest is the extent prospective entrants are familiar with other incumbent exporters already exporting to the market. More concretely, MMC_{ict} is the frequency with which firm i meets all rivals in country c in all other countries at time t. For example $MMC_{ict} = 2$, indicates that firm i faces incumbents in country twice in other markets.[9] Figure 4 provides an example of how the MMC measure is calculated for a hypothetical firm.

Illustration of MMC as a Spillover Measure

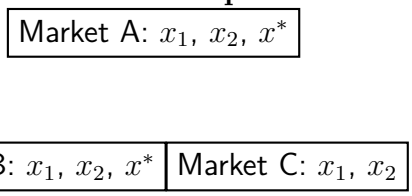


Figure 1: Firms x_1, x_2 and x^* export to markets A and B and x^* is considering exporting to market C. $MMC_{1C} = 4$. To see this, x^* meets x_1 twice in markets A and B (2 contacts) and x_2 twice in the same markets (another 2 contacts) leading to 4 total contacts.

I control for other variables similar to the extensive margin model used in Chaney (2014) to ascertain whether MMC could be a valid gravity model. $|C_{it}|$ is the number of destinations

[6] See Larch et al. (2025) for a detailed overview of this dataset.

[7] See Gurevich and Herman (2018) for a detailed overview of this database.

[8] Details of the empirical approach are discussed in the Section 4 below

[9] This is a similar measure to that used in Boohaker (2025). Note that one possibility is that there are two incumbents in country c that i faces in one other country, one incumbent that the firm faces in 2 other markets or 2 incumbents faced in two different countries.

served by firm i, and N_{ct} is the number of incumbent U.S. exporters in country c. The variables d_{ict} and b_{ict} measure the average distance and average battery imports between country c and the firm's existing export destinations. Equation 2 shows the computation for d_{ict} and equation 3 shows the calculation for b_{ict}.

$$d_{ict} = \frac{\sum_{c' \in C_{it}} d_{c',c}}{|C_{it}|} \tag{2}$$

$$b_{ict} = \frac{\sum_{c' \in B_{it}} B_{c'ct}}{|C_{it}|} \tag{3}$$

Additional controls include distance from the U.S., total U.S. battery exports to c, country remoteness measures ($\bar{d}_c, \bar{b}_{ct}$), firm employment, total export value, and the share of related-party exports. Year fixed effects are included, and standard errors are clustered at the firm level.

5 Results

MMC is a significant factor in U.S. battery manufacturers export decision making. Table 1 reports the extensive-margin results. A 10 percent increase in MMC raises the probability of export entry by approximately .01 percentage point, or a .36% increase relative to a baseline export entry rate of 3%. Other gravity related variables such as distance (d_{ict}) and level of imports (b_{ict}) from incumbent countries had effects consistent with theoretical expectations: shorter distance and more exports from incumbent markets had positive effects on entry decisions that were similar in magnitude to Chaney (2014). Excluding MMC leads to noticeable changes in the estimated coefficients on standard gravity variables, consistent with omitted variable bias. Standard gravity relationships of distance and import volume hold true. Magnitudes to the other gravity variables are also similar to the reduced form estimates displayed in Chaney (2014). Unsurprisingly, more export countries for the firm ($|C_{it}|$) and more U.S. battery exporters in the country (N_{ct}) are associated with a higher likelihood of firm i exporting to the country in the next period, which is consistent with Kamal and Sundaram (2016). Moreover, excluding MMC proves to be a source of omitted variable bias for other commonly used gravity related variables in predicting export decisions such as distance and the level of imports for these firms. Specifically, including MMC suggests the effect of distance is biased downward while the effect of exports is overstated for these firms.

Table 2 reports intensive-margin results. A 10 percent increase in MMC is associated with roughly 2.4 percent higher export value. Consistent with gravity predictions, export value falls with a market's average distance to the firm's other destinations and rises with its imports from those destinations. Export value falls if other export markets for the firm are farther away, but increase if export values from those markets are higher, which is consistent with predictions from standard gravity models. Furthermore, the coefficient magnitude of marginal changes in MMC is comparable to marginal changes in these measures as well. These estimates are conditional correlations, not causal effects. MMC is built from rivals'

Extensive Margin: Export Entry and Multimarket Contact

		Baseline (b/se)	MMC (b/se)
$\|C_{it}\|$	Number of Export Countries for i	0.00018***	0.00015***
N_{ct}	Number of U.S. Exporters in c	0.00001***	0.00001***
d_{ict}	Average Distance from Incumbent Markets	-0.00120***	-0.00157***
b_{ict}	Average Exports from Incumbent Markets	0.00045***	0.00038***
$b_{US,c}$	Exports from U.S.	0.00142***	0.00136***
X_{it}	Total Firm Exports	0.00098***	0.00083***
$d_{US,c}$	Distance from U.S.	5.00×10^{-5}	0.00022*
P_{ct}	Total Production and Exports form c	0.00006***	0.00005***
MMC_{ict}	Log Multi-Market Contact		0.00107***
R^2	Pseudo R-squared	0.541	0.543

Table 1: Coefficients are marginal effects computed at the average of each independent variable. There are roughly 2,900 firms in the sample. Sample includes all U.S. exporters with a manufacturing establishment that export rechargeable batteries. The following variables are included in the regression but are not reported in the table to comply with Census Bureau guidance on disclosure-avoidance protection: E_{ict} (whether firm i exported to country c at time t) in addition to the size of the firm in terms of employment and number of establishments, the share of export shipments that are to related parties, the average distance from other countries $\bar{d}_c$ and total global battery imports to the country ($\bar{b}_{ct}$). Standard errors are clustered at the firm level and all are less than 0.0001. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

market choices, so common shocks and sorting (the reflection problem, Manski, 1993) cannot be ruled out.

Table 3 shows summary statistics around the sample of firms and provides additional information on the export activity of these firms from 2016 - 2019. There are roughly 2,900 firms across 1,423,000 total observations.[10] The extensive margin of trade at the firm-country level is roughly 3 percent. There are 25 U.S. exporters servicing the average country in the average year, and the average firm exports to roughly 4 markets in the average year. The average firm looking to export to the average country in the average year is expected to meet the incumbent firms in over 260 other instances. Given the heavy skewness of this statistic, it is prudent to consider log normalized changes in this variable in the regression analysis. The average firm in this sample exports roughly $2.5 million worth of rechargeable batteries in the average year to all its destination countries.

[10]Since we want to study whether exports choose to export to a country in the next period, the analysis data set consists of all possible firm-country pairs in each period.

Intensive Margin: Export Value and Multimarket Contact

		(1) Baseline	(2) Including MMC
d_{ict}	Average Distance from Incumbent Markets	-0.3008^{***}	-0.3806^{***}
		[0.072]	[0.073]
b_{ict}	Average Exports from Incumbent Markets	0.1432^{***}	0.1247^{***}
		[0.017]	[0.017]
$log(MMC_{ict})$	Log Multi-Market Contact		0.2535^{***}
			[0.049]
R^2		0.308	0.310

Table 2: There are roughly 2,900 firms in the sample. Sample includes all U.S. exporters with a manufacturing establishment that export rechargeable batteries. Intensive margin measures export value of firms to country c, which differs from extensive margin that measures whether exports occur in the following period. I suppressed results that are not key to motivating the main result since Census limits the output in proportion to the number of firms included in the sample. The following variables are included in the regression but are not reported in the table to comply with Census Bureau guidance on disclosure-avoidance protection: E_{ict} (whether firm i exported to country c at time t) in addition to the size of the firm in terms of employment and number of establishments, the share of export shipments that are to related parties, the average distance from other countries $\bar{d}_c$ and total global battery imports to the country ($\bar{b}_{ct}$). Estimates for $|C_{it}|$, N_{ct}, $d_{US,c}$ and $b_{US,c}$ seen in Table 1 are also accounted for, but suppressed in this table. Standard errors in brackets and are clustered at the firm level. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Summary Statistics

Variable	Description	Mean	Std. Dev.
Export entry ($E_{ic,t+1}$)	1 if i exports to c at time t+1	0.03	0.15
$\log(MMC_{ict})$	Multi-market contact measure (logged)	5.0	1.0
MMC_{ict}	Multi-market Contact measure	268	377
$\|C_{it}\|$	# of export markets for firm i	4.0	8.0
N_{ct}	# of U.S. battery exporters to country c at time t	25	48
X_{it}	Firm Export value of batteries	\$2,557,000	\$18,830,000
Firms	2,900		
Observations	1,423,000		

Table 3: Number of observations and firms are rounded to 4 significant digits to comply with Census bureau guidance on disclosure-avoidance protection.

6 Conclusion

This paper provides new evidence that exporters' interactions in foreign markets shape international expansion in economically meaningful ways. By introducing multimarket contact as a measure of familiarity among exporters abroad, the analysis extends the export spillover literature beyond domestic geography and industry linkages, highlighting foreign markets

themselves as sites of learning and information transmission. The empirical results show that firms are more likely to enter a destination—and to export more once there—when they have greater prior overlap with incumbent exporters in other markets. These effects are large relative to standard gravity forces, underscoring the importance of exporter networks formed outside the home country.

The findings contribute to the multimarket contact literature by documenting a context in which MMC promotes, rather than deters, entry. While earlier work often emphasizes mutual forbearance and entry deterrence, the results here suggest that in industries characterized by high fixed costs, regulatory complexity, and product differentiation, MMC can facilitate entry through learning, coordination, and improved access to foreign intermediaries. The theoretical extension clarifies this mechanism by showing how learning-induced reductions in fixed costs and higher markups can outweigh the threat of cross-market retaliation.

From a policy perspective, the results suggest that export promotion efforts that foster repeated interaction among exporters—such as trade missions, industry consortia, and foreign market networking events—may generate spillovers even among competitors. For firms, the findings imply that observing and engaging with rivals abroad can be a strategic complement to market entry rather than a deterrent.

Several limitations point to avenues for future research. The analysis focuses on a single industry and country, and the strength of MMC effects may vary across sectors with different competitive structures or degrees of product differentiation. Future work could extend the framework across industries, explore dynamic adjustments over longer horizons, incorporate direct measures of firm-to-firm relationships, or extend the theoretical model to allow for a structural estimation of the MMC effect. More broadly, the results suggest that incorporating multimarket interaction into models of firm-level trade can yield a richer understanding of how exporters learn, compete, and expand globally.

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